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ORIGINAL INVESTIGATION

Effects of Prematch Travel and Training on Physical and Technical Performance in International Football Matches

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Purpose: This study examined the relationship of prematch travel and in-camp training on ensuing physical and technical match performance of footballers (soccer) competing for a national team. ***Methods:*** Match running and technical performance data were obtained from 68 national-team footballers competing in international matches (N = 108). Match performance data were aligned with the confirmed travel durations, time-zone change, travel direction, and time between arrival and kickoff for the travel to the match. In addition, in-camp training load from the 3 days prior to national-team matches was also collated. Linear mixed models assessed relationships between travel and training measures with physical and technical match performance outcomes. ***Results:*** Travel variables explained little variance in outcomes (R^2 = .02–.16). Traveling eastward was associated with an increase in total (P = .042) and very high-speed distance (P = .030) in matches and a 5% decrease in pass accuracy (P = .012). Greater time-zone difference was associated with increased match decelerations (P = .027), while arriving later for matches was associated with increased total (P = .041) tackles. Increases in training high-speed distance were associated with increases in match high-speed distance (P = .004) and number of decelerations (P < .001). An increase in training total distance was associated with a decrease in match decelerations (P = .002). ***Conclusions:*** Prematch travel appeared to have minimal effects on physical and technical performance in this national team. Training loads prior to matches may have some relationship with match running performance, and, thus, readiness to train should be a priority for athletes arriving in national-team camps.

Keywords: jet lag, national team, match performance, travel fatigue

International football (soccer) matches require selected players to undertake travel directly from clubs to national-team camps, often arriving with only a few days to recover and prepare for the match.[1] Such situations can place significant stress on the athlete and may risk reduced physical and technical performance when playing for their national team and may also occur for some club teams.[2] These performance concerns relate to the potential for increased physical and mental fatigue caused by jet lag, travel fatigue, and disrupted sleep that are common following extensive travel.[3] Additionally, with often only 2 to 5 days of camp to prepare for matches, training and preparation following travel to national-team matches is challenging.[4] Further understanding of the relationships between travel to national-team matches and training load upon arrival with physical and technical match performance can assist athlete arrival plans for both club and national teams.

Given the short time to recover between arrival and competition for national-team footballers, elevated fatigue and impaired sleep as a result of jet lag and travel fatigue may reduce an athlete's physical performance.[3] Previous studies in international football tournaments have observed varying effects of travel on match running, with reduced total distance (TD) covered in matches following eastward travel observed during the 2018 FIFA World Cup in Russia,[5] though no effects of travel distance on TD per minute, sprints per minute, and high-speed distance per minute were observed in the 2014 FIFA World Cup in Brazil.[6] Such studies, however, only focus on the final stages of these major tournaments, where players can arrive weeks in advance of matches, and prematch travel demands are short. Qualifying stages for tournaments are often played within the domestic club season and, as such, athletes are required to travel directly from club to national team, with these commitments delaying athlete arrival to only 2 to 5 days prior to matches. Studies assessing the effects of differing travel demands in international Rugby Sevens have identified reduced TD and distance >5 m/s during tournament matches after long-haul (>5 h) compared with short-haul (<5 h) travel.[7] By contrast, no effect of travel duration or number of time zones crossed on running performance was reported in other research on international Rugby Sevens athletes.[8] Neither study, however, reported the time between arrival and matches, and as such the effect of this remains unclear. Given the conflicting findings and lack of studies examining travel directly into international football (soccer) matches, it is still unclear how travel into a match will affect physical match running performance.

Extensive travel into a national team may reduce match technical skill execution due to impairments in cognitive performance associated with jet lag, travel fatigue, and sleep disruption.[9] Despite this, no studies have explored the effects of varying prematch travel demands on technical performance in football. Fowler et al[10] observed fewer goals conceded, more shots on goal and corners, and fewer opposition shots on goal and corners in home matches compared with away matches in the Australian domestic football (soccer) league. However, travel demands for such matches are relatively short (5.3 [2.2] h), and given the similar travel demands across all players, it is difficult to separate out effects of travel from those of away match-disadvantage. Research in Rugby Sevens athletes identified significant reductions in several technical performance indicators such as, passes, carries, clean breaks, defenders beaten, and meters gained, following both

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eastward and westward travel (24 h, 12 time zones), with a more significant reduction from eastward trips and independent of away-match disadvantage.[11] Hence, further research is needed to determine the influence of prematch travel demands on technical performance in national football teams.

Following travel to national-team camp, athletes are required to train for 2 to 5 days in preparation for the matches while recovering from the potential effects of travel. The combined influence of both travel and training stresses may induce additional fatigue and influence ensuing physical and technical match performance. Studies in club-based environments report positive relationships between running loads in training and the subsequent running performance in matches, though they are independent of travel demands.[12–15] In addition, despite the potential for both physical[16] and mental[17] fatigue to affect technical performance, the combined effects of travel and training prior to matches on technical performance has not been explored. Accordingly, this study comprises 2 aims: first, to identify the effect of the travel duration, direction, time zone change, and time between arrival and the match on measures of physical and technical performance in national-team matches, and second, to assess the influence of measures of external training load 3 days prior to matches on physical and technical match performance in national-team footballers.

Methods

Participants

Participants were players competing for a senior male international football (soccer) team between October 2018 and March 2023. During this period, the team competed in qualifying matches for the 2022 FIFA World Cup and friendly matches after the tournament. Data were collected from 68 athletes (mean age 25.4 [3.6] y). Through individual contracts with the national football federation, athletes and the National Federation consented to the use of their anonymous data for research purposes. Ethical approval was obtained from the institutional human ethics committee (ETH22-7708).

Overview

Data were collected from players involved in each national-team match during the study period (n = 308). Goalkeepers, players who played <60 minutes in the match, or players who arrived >7 days before the match were excluded from the final data set (n = 103). Both physical and technical match performance data were obtained for each player. Match performance data were aligned with their travel demands into the match and training load for 3 days prior to the match. To account for effects of opposition quality, the FIFA ranking of each match opponent was obtained from a historic database at the time of the match. Start times for all matches were between 18:00 hours and 21:30 hours local time.

Training and Match Physical Data

Physical running data for matches and training were obtained using global positioning system (GPS) units (STATSports, Apex) worn between the players' scapulae using a customized vest. Each unit includes a gyroscope (10 Hz), triaxial accelerometer (100 Hz), and magnetometer (10 Hz). These devices have previously been validated for the measurement of speed and distance variables in team sports.[18,19] All GPS units were managed by the sport scientist in camp, with all data downloaded and processed using the manufacturers software package (STATSports Sonra, Apex). For both training and matches, the following variables were obtained: TD, total high-speed distance (19.8–25.2 $km \cdot h^{-1}$), total very high-speed distance (>25.2 $km \cdot h^{-1}$), number of accelerations (>3 $m \cdot s^{-2}$), and number of decelerations (>3 $m \cdot s^{-2}$). These thresholds had been previously used by the national team based on FIFA guidelines and their frequent use in professional football.[20,21] For matches, each variable was converted to a relative measure by dividing the raw value by the number of minutes the athlete played in the match. To quantify total training load prior to matches, the sum of the total values for each day for 3 days prior to matches was calculated for each GPS metric.[20] A 3-day sum was used as this best represented the amount of training days that most players took part in upon arrival into the national team. Data points in which players elected not to wear a GPS unit (n = 25) were excluded from the data set for use in physical performance models (final data set, n = 79). In addition, due to a change in acceleration threshold during the study period and unavailability of data prior to this change, data points prior to the change were not included in the match accelerations model (n = 35).

Match Technical Data

Technical data from all national-team matches were collated from a match statistics partner (Opta Data, Stats Perform). The system used to measure technical data has been previously observed to have high interoperator reliability (kappa value = .92–.94).[22] A list of all variables obtained and their definitions can be found in Table 1. These variables were selected based on their commonality in a review of previous literature.[23–25] Technical match data were not available for one match and thus were excluded from analysis (final data set, n = 95).

Travel Details

Travel details of all flights prior to matches were obtained from booked travel schedules with details of the flight and scheduled arrival and departure times. The actual arrival and departure times of each flight were verified via an online flight database (flightera.com). The total travel duration was calculated as the total amount of time between aircraft departure and arrival in the final destination (including the length of stopovers). The direction of travel was determined based on geographical locations of the cities of arrival and departure and labeled as either east or west. The time zone difference was calculated as the difference in time zone between the departure city and the arrival city at the time of arrival. The number of hours between arrival from travel and the national-team match kickoff was also included for analysis. These variables have been previously observed to influence perceptual fatigue, jet lag, and sleep ratings and thus may represent travel stress of the athlete.[26]

Statistical Analysis

Match physical and technical performance measures were aligned with the travel and training load performed by the player prior to each match. All data were imported into R Studio for analysis. For each match running outcome variable, linear mixed models were built using the lme4 software package.[27] To account for nonindependence of the observations, an anonymous player code and the match ID (which accounts for the time of match and opposition) were included in the model as random effects. In addition, the playing position was included as a random effect to account for positional differences in performance outcomes. Playing positions were categorized as Fullback, Central Defender, Central Midfielder,

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Table 1 Operational Definitions for Technical Performance Statistics Derived by Opta

Outcome variable	Definition
Total touches	A sum of all events where a player touches the ball, so excludes things like Aerial lost or Challenge lost.
Successful touches	Calculated as the difference between total touches and unsuccessful touches.
Unsuccessful touches	When the ball bounces off a player and there is no intentional pass, we award a touch. When a player miscontrols the ball with a poor touch, we award an unsuccessful touch. Also used for mishit shots that go backward toward a player's own goal.
Total passes	Any intentional played ball from one player to another. Passes include open play passes, goal kicks, corners, and free kicks played as pass—but exclude crosses, keeper throws, and throw-ins.
Accurate passes	A completed pass is a pass that goes to a teammate directly without a touch from an opposition player.
Inaccurate passes	Calculated based on the difference between total passes and accurate passes.
Pass accuracy, %	Calculated as the percentage of accurate passes to total passes.
Total shots	Calculated as the sum of shots on target and shots off target.
Shots on target	A shot on target is defined as any goal attempt that: • Goes into the net regardless of intent—For Goals only. • Is a clear attempt to score that would have gone into the net but for being saved by the goalkeeper or is stopped by a player who is the last-man with the goalkeeper having no chance of preventing the goal (last line block).
Shots off target	A shot off target is defined as any clear attempt to score that: • Goes over or wide of the goal without making contact with another player. • Would have gone over or wide of the goal but for being stopped by a goalkeeper's save or by an outfield player. • Directly hits the frame of the goal and a goal is not scored.
Total crosses	Any intentional played ball from a wide position intending to reach a teammate in a specific area in front of the goal.
Accurate crosses	An accurate cross is a cross that goes directly to a teammate without a touch from an opposition player.
Inaccurate crosses	Calculated as the difference between total crosses and accurate crosses.
Total tackles	A tackle is defined as where a player connects with the ball in a ground challenge where he successfully takes the ball away from the player in possession.
Won tackles	A tackle won is deemed to be where the tackler or one of their teammates regains possession as a result of the challenge or that the ball goes out of play and is "safe."
Interceptions	When a player intercepts any pass event between opposition players and prevents the ball reaching its target. Cannot be a clearance.
Clearances	This is a defensive action where a player kicks the ball away from their own goal with no intended recipient.
Turnovers	Loss of the ball through a mistake/poor control of the ball.

Wide Midfielder, or Forward. Models were built using a forward stepwise approach with the fit of each fixed effect assessed through its statistical significance, the Aikake Information Criterion value, and the R^2 values of the model. Assumptions of normality and homogeneity of variance were assessed post hoc using Q-Q plots and residual plots, and no issues were identified.

Technical performance (count data) outcomes were analyzed using generalized linear mixed models with a Poisson link function. To account for differences in the number of minutes played in each match, technical outcomes were converted to rates (number of events per minute of play) by including the minutes played as an offset within the models. For counts that included large numbers that were not bound by zero and data that represented a normal distribution, linear mixed models were used (total touches, successful touches, accurate passes, inaccurate passes, and pass accuracy). The random effects of player code, match ID, and match position were included in all models.

Results

Travel and Training Loads

To provide context to prematch travel demands in national-team footballers, the distribution of flights by travel duration and the time between arrival and match start are shown in Figure 1. Travel ranged from 0.85 hours to 31 hours duration (mean: 13.4 h [9.6 h]), across 0 to 11 time zones (mean 5.2 [4.3]) and included both east (N = 75) and west (N = 29) directions. Mean travel durations for each travel direction were 16.6 (8.4) hours for eastward travel and 5.5 (8.0) hours for westward travel. Athletes arrived on average 84.8 (29.8) hours prior to the match and tended to arrive earlier when longer travel durations were required, with very few arrivals <40 hours before a match (95% of athletes arrived at least 48 h prior to matches). The mean FIFA ranking of opposition teams was 93 (45) and was lower (ie, better teams) for trips involving eastward travel (76 [35]) than westward travel (135 [39]). Mean training loads over the first 3 days of arrival were 8701.5 (2671.5) m TD, 314.7 (173.7) m high-speed distance, and 59.9 (173.7) m very high-speed distance. Mean accelerations (>3 $m \cdot s^2$) and decelerations were 89.5 (34.7) and 66.6 (28.7), respectively.

Physical Match Performance

Final model outputs and details of significant explanatory variables for physical match performance are shown in Table 2. Model R^2 conditional values were relatively high ($R^2 = .84–.89$). However, R^2 marginal values were minimal ($R^2 = .02–.16$), indicating that the random effects included within each model explained the majority of variance in each outcome. Athletes traveling eastward covered 3.003 m per minute more in TD ($P = .042$) and 0.336 m per minute

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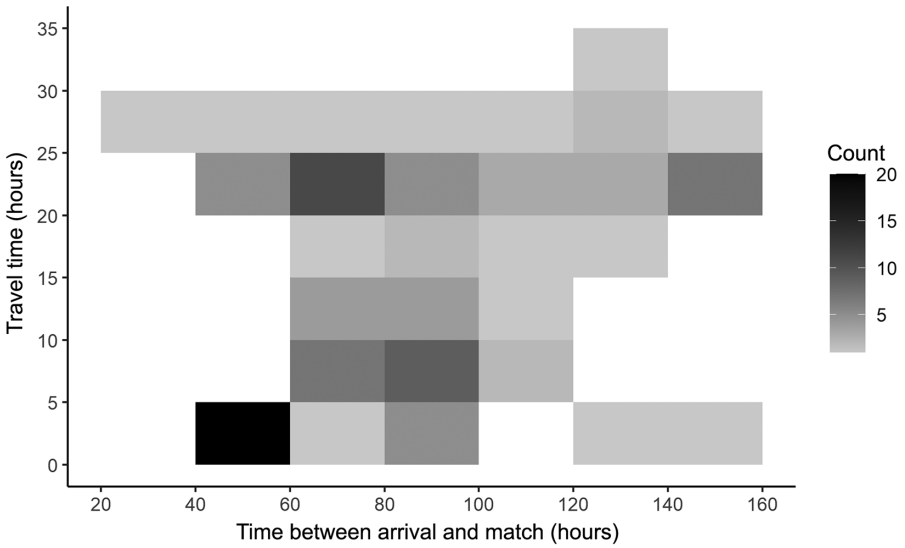

Figure 1 — Distribution of flights based on the players' travel time and when they arrived relative to the match start time (N = 103).

Table 2 Final Model Outputs for Significant Variables Explaining Match Global Positioning System Measures

Outcome	R^2 conditional	R^2 marginal	Explanatory variables	Coefficient	*P*
Match TD, m/min	.84	.03	Traveling eastward	3.003	.042
Match HSD, m/min	.89	.03	Training HSD, m	0.002	.004
Match VHSD, m/min	.85	.02	Traveling eastward	0.336	.030
Match accelerations per minute	.83	.00	—	—	—
Match decelerations per minute	.86	.16	Training HSD, m	0.0006	<.001
			TD, km	–0.028	.002
			Time-zone difference, h	0.008	.027

Abbreviations: HSD, high-speed distance; TD, total distance; VHSD, very-high-speed distance. Note: All models included the player ID, match ID, and player position as random effects. Coefficients (and direction) for numerical explanatory variables represent the change in outcome per 1 unit change in explanatory variable (units are given).

more in very high-speed distance ($P = .030$) compared with westward travelers. For each additional time zone crossed during travel, there was a significant increase of 0.008 decelerations per minute ($P = .027$). No other travel variables showed significant relationships with match running outcomes ($P > .05$).

An influence of in-camp training loads was evident, with a 1 m increase in training high-speed distance associated with a 0.002 m per minute increase in match high-speed distance per minute ($P = .004$) and a 0.0006 increase in the number of decelerations per minute ($P < .001$). A 1 km increase in training TD was associated with a 0.028 decrease in decelerations per minute in the match ($P = .002$).

Technical Match Performance

Final model outputs for technical performance outcomes are shown in Table 3. Only pass accuracy and total tackles had a significant relationship to any travel-based explanatory variables, and R^2 values were small ($R^2 = .07$ and $R^2 < .01$). Traveling eastward was associated with 5% reduction in pass accuracy compared with athletes traveling westward to matches ($P = .012$). Reductions in total tackles per minute were associated with arriving earlier to the match. For every hour prior to the match that an athlete arrived, total tackles decreased by 0.7%.

Discussion

This study explored the effects of prematch travel and in-camp training on ensuing physical and technical performance in international football matches. The effect of travel duration, direction, and time between arrival and matches on physical and technical performance measures was minimal. In addition, a small relationship between training load upon arrival and high-speed running quantities in matches was observed. Although prematch travel may induce fatigue and poor sleep,[26] normal match factors, such as opposition quality,[28] are likely to have had larger influence on physical and technical performance than travel demands. Furthermore, for athletes arriving at least 48 hours prior to national-team matches, travel did not appear to negatively impact match performance in this national team.

Small associations ($R^2 = .02$–$.16$) between travel direction and match physical performance measures were observed in this study, with athletes traveling eastward covering a greater amount of total and very high-speed distance in matches compared with when traveling westward. Although differences in physical performance may be expected based on travel demands, eastward travel would be expected to have a more negative effect on match performance than westward.[29] These conflicting findings may therefore be better

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Table 3 Final Model Outputs for Variables Explaining Match Technical Performance Measures

Outcome	R^2 conditional	R^2 marginal	Explanatory variables	Change from intercept, %	*P*
Total touches	.75	0			
Successful touches	.77	0			
Unsuccessful touches[a]	.43	0			
Total passes	.80	0			
Accurate passes	.80	0			
Inaccurate passes	.32	0			
Pass accuracy, %	.27	.07	Traveling eastward	↓ 5%	.0117
Total shots[a]	.80	0			
Shots on target[a]	.27	0			
Shots off target[a]	.12	0			
Total crosses[a]	.72	0			
Accurate crosses[a]	.53	0			
Inaccurate crosses[a]	.66	0			
Total tackles[a]	.02	<.01	Time between arrival and match	↓ 0.7% per 1 h	.0409
Won tackles[a]	.05	0			
Interceptions[a]	.40	0			
Clearances[a]	.40	0			
Turnovers[a]	.43	0			

Note: All outcome variables are expressed as relative measures per minute of match play. All models included the player ID, match ID, and player position as random effects.
[a]Use of a Poisson regression model.

explained by competition scheduling and location factors as athletes traveled eastward into matches against higher quality teams (based on FIFA rankings). For example, the FIFA ranking of opponents following eastward trips was 49.5 points lower (ie, better teams) than that of westward trips. This aligns with prior club-based research in which increases in match running variables have been observed when competing against better quality opposition.[28] Although it is unlikely that full circadian resynchronization would have occurred in the athletes in this study,[3] the lack of effects on physical performance could result from 95% of athletes arriving at least 48 hours in advance of matches and thus having time to recover from travel.[3] As such, it could be recommended, where possible, that athletes arrive a minimum of 48 hours prior to national-team matches (or longer for greater travel demands) to minimize the risk of detrimental effects on performance from prior travel demands.

Although prematch travel had a minimal effect on physical match performance in this cohort, hastening the availability to train may be important as small associations existed between training loads on arrival and subsequent match running outcomes. A small positive relationship was observed between the total high-speed distance completed in the 3 days prior to a match and the high-speed distance ($R^2 = .03$) and number of decelerations performed in the match ($R^2 = .16$). Previous club-based research has observed similar positive relationships between prematch training and match running variables,[12–14] whereas reduced match acceleration performance has also been linked to excessively high or low acute sprint loads in training.[15] Although previously this relationship has been related to a fitness–fatigue effect,[15] the variations in training running loads in the current study may reflect adjustments in the tactical plans for the upcoming match. Teams are likely to attempt to train for the tactics that they use in a match, and thus position-based match tactics that require greater running loads may result in increased running loads during training. An additional finding of this study was that increases in TD in training were related to reductions in match decelerations. Although the reason for this is unclear, it may be related to the nature of the distance covered, with players covering greater distances tending to do so at lower speeds. Regardless, these findings highlight the importance of athlete preparedness to train and compete in the congested schedule of a national team, and hence some need to promote recovery from travel exists.

For the cohort in this study, prematch travel demands and training loads upon arrival appeared to have minimal relationships with technical match performance. Traveling eastward to a match was associated with a small reduction in pass accuracy ($R^2 = .07$) when compared with traveling westward. Again, for athletes in this cohort, eastward travel required longer travel durations and potentially worse jet lag symptoms;[29] however, this finding may also be related to the previously discussed increased opposition quality following eastward trips. This is consistent with prior research that observed reductions in pass accuracy being associated with increased opposition quality in international youth football.[25] Associations were also observed between time between arrival and match and the total number of tackles, whereby players who arrived later (closer to the match) had more total tackles. However, as indicated by a marginal R^2 of less than .01, the strength of this association was minimal. As such, the findings of this study suggest match technical performance for this national team was unlikely to be affected by travel or training demands for athletes arriving at least 48 hours prior to matches.

Despite the novelty of this data to inform prematch travel and training in national teams, there are several limitations that should be considered when interpreting these results. First, these findings represent only a single national team, and thus variation in outcomes may be expected across other teams who may have different travel demands or strategies surrounding travel. Furthermore, due to the need to play at least 60 minutes in each match and the unavailability of GPS data for some athletes, a relatively small sample size of 103 data points was used. In additional, only 2 of the 14 matches involved in this study were against higher ranked

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teams, and most matches resulted in a win (71%). As such, impacts of travel and training on performance may have been masked by the superior quality of the analyzed team compared with their opponents. Finally, limited control was permitted over what athletes did before, during, or after travel, and thus the use of jet lag management strategies was not controlled for, which may influence responses to travel.

Practical Applications

- Travel to international football matches had small effects on physical or technical performance in this national team, and thus athletes should aim to arrive at least 40 hours in advance of matches.
- Small relationships existed between total training loads for 3 days prior to a match and match running performance, and thus readiness to train should be a priority for athletes arriving at national-team matches.

Conclusions

In conclusion, this study examined the relationship of prematch travel on the technical and physical performance of footballers competing in a national team. These findings show that travel to matches had only small effects on physical or technical match performance, with most of these athletes arriving at least 48 hours prior to matches. When earlier arrival is not possible and travel durations are long, particular focus on interventions to assist the alleviation of jet lag and manage sleep deficits could become more important. In additional, small relationships existed between total and high-speed running in training and high-speed running in matches, thus suggesting that readiness to train should be a priority for players arriving at national-team matches. Future studies should seek to add to these findings using other population groups and travel demands and explore the effects of travel in which arrival at least 48 hours prior to matches is not possible.

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